PROJECT PROPOSAL

DS-GA 3001 Special Topics:
Data Visualization for ML

What Makes The Cut?

Interpretable Shot Selection and Clustering of Movie Trailers

Team 13
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1. Research Background and Related Work

1.1 Key Paper

Argaw, Soldan, Pardo, Zhao, Caba Heilbron, Chung, Ghanem. Towards Automated Movie Trailer Generation. CVPR 2024. https://arxiv.org/abs/2404.03477.

1.2 Summary

The authors cut films and trailers into shots with TransNet V2, and each shot is embedded with CLIP ViT-H/14. Their Trailer Generation Transformer (TGT) has three parts: a one-layer trailerness encoder that scores every film shot, a four-layer context encoder over the whole film, and a five-layer decoder that predicts the next trailer shot, snapped to the nearest film shot. TGT is trained on 23,304 film-trailer pairs and reaches F1 shot selection of 46.8 on the MovieNet benchmark of 989 films, compared to 39.3 for CLIP-It and 5.3 for random. Adding an IMDb plot summary pushes F1 to 53.7.

1.3 Main Contributions

The first model that learns both which shots a trailer uses and in what order, an explicit per-shot trailerness score, and MovieNet as a trailer benchmark.

1.4 Issues with the Paper

1. The model is a black box. Their transformer does not explain why a shot is chosen. There is no ablation by cinematography, dialogue, or story position.
2. Narrative position is unused. Plot text was used to raise F1. The paper never asks how far into the story a trailer reaches, or how much it spoils the movie.

1.5 What We Aim to Build On and Extend

We want to build on their *trailerness* methodology, CLIP shot embeddings, using MovieNet as the benchmark, and precision and F1 at trailer length as metrics.

We seek to extend it in three ways:

- Build and measure shot alignment. The paper did not measure this extensively, mostly relying on CLIP similarity.
- Replace the black-box decoder with interpretable models, showing factors that lead to selection.

- Move from single shots to whole trailers, clustering them by pacing and story sampling, following Fish et al., who clustered trailers on audio-visual content: https://arxiv.org/pdf/2012.02639.

2. Research Questions and Proposed Contributions

When a trailer for a film is made, what predicts which shots get used, how deep into the story they reach, and do trailers fall into recognizable groups?

2.1 Research Questions

RQ1, Selection and Story Depth. Can a model predict which film shots appear in the trailer from visual content, shot scale, dialogue, and story position? Does story position add predictive power beyond content (feature ablation and SHAP), or is trailerness mostly about how a shot looks? Using the matched shots, how deep into the film do trailers reach, measured by share of shots from the final act?

RQ2, Clustering Analysis. Do trailers share similar pacing, audio characteristics, and dialogue sentiment patterns? We cluster trailers on how they are paced and how they sample the film, then test whether the clusters follow genre, era, or neither.

2.2 Stretch Goals

S1, Spoiler Content. An LLM rates each synopsis paragraph from 0 (setup) to 5 (major reveal), so we can measure how often trailer shots come from reveals rather than just late scenes.

S2, VLM Spoiler Detector. A vision-language model receives a keyframe and the film's synopsis and judges whether the shot reveals a major plot event, with calibration checked by era and genre.

2.3 Proposed Contributions

1. An open trailer-to-film shot alignment dataset, visualized.
2. An interpretable trailerness model with ablations and SHAP attributions.
3. A trailer clustering on pacing, audio, dialogue, and story sampling, with comparison against genre and era.

3. Methods and Visualization Design

3.1 Connection to Visualization and ML

RQ1's classifier produces per-shot scores that are not interpretable. We need to see which shots it ranks high that editors skipped, and why. Visualization is what helps show this best.

RQ2's clusters make more sense when we plot the exact groupings. The approaches below are designed around those clustering outputs, similar to Calibrate and Neo learned in class.

3.2 Proposed Approach

1. Shot Alignment

Detect trailer shots with TransNet V2, embed trailer frames and MovieNet film keyframes with CLIP, and match each trailer shot to its most similar film shot. Matched film shots become our *trailerness* labels.

2. Trailerness Model (RQ1)

One row per film shot, with features for CLIP embedding, shot scale, dialogue, story position, and shot length. We will fit logistic regression, Random Forest, and LightGBM (or other explainable models), and explain them with SHAP, dropping one feature group at a time to assess the impact of story position within the trailer.

3. Clustering (RQ2)

Cluster based on a combination of pacing, audio energy, dialogue sentiment, and story sampling. We will cluster using HDBSCAN and k-means (possibly more methods) in the original feature space, and repeat this per modality.

3.3 Visualization Design

We propose building three linked D3 views:

View	**Design and Purpose**
Overview	PCA/UMAP/t-SNE scatter of trailers, colored by cluster, genre, decade/era, etc.
Cluster View	The selected cluster's defining features compared with the full corpus (e.g. shorter shots, higher audio energy).
Trailer View	One trailer's synchronized timeline (cuts, audio energy, pacing), with links from each shot to its source in the film. Film shots are colored by trailerness and show their SHAP explanation.

4. Data, Evaluation, and Feasibility

4.1 Data

We will use MovieNet (Huang et al. 2020), which provides keyframes for 1,100 films, synopsis-to-shot alignment for 328 films, metadata, subtitles, and YouTube IDs for trailers, which we will download via the `yt-dlp` package. We verified this from the dataset. If necessary, we can also use Trailers12k and Cinescale for a richer trailer/movie set.

4.2 Implementation and Literature

The pipeline will be in Python: TransNet V2, CLIP, LightGBM, SHAP, librosa, HDBSCAN, and PCA/UMAP/t-SNE, and the views will be in D3. We aim to build on Argaw et al. (2024) for *trailerness*, Fish et al. (2021) for trailer clustering, and Calibrate and Neo for model-diagnosis visualization.

4.3 Evaluation

RQ1: Shot Alignment and Trailerness Model

- **Alignment**: Accuracy at recovering the source shot of film shots edited to look like trailer footage (crops, color grading, text overlays). Target: at least 0.9.
- **Trailerness model**: AUPRC, precision and F1 at trailer length, with cross-validation grouped by film. Baselines: random, position only, and CLIP only, similar approach to the key paper.
- **Ablations**: How much AUPRC drops when each feature group is removed. This showcases the impact of story position.

RQ2: Clustering Analysis

- **Cluster quality**: Use clustering metrics (e.g. silhouette score) to assess cluster cohesion and separation in the original feature space, alongside qualitative inspection and stability checks.
- **Modality comparison**: Compare each modality clustered alone with all combined to explore how each modality influences the groupings.
- **Cluster characteristics**: Agreement of clusters with genre and decade. Examine whether each cluster's automatically generated feature summary accurately describes its observed characteristics.

Visualization Evaluation

Case studies showing the tool supports three tasks:

- Explaining why a trailer is an outlier from its cluster signature.
- Relating a change in audio volume to a shift in dialogue sentiment.
- Tracing where a trailer's shots come from and why the model picked a shot.

We will also handle overplotting in the overview visualization with tooltips, highlighting, and zoom.

5. Project Timeline

Dates	**Milestones**
October 20 – November 3	Data exploration, scraping, alignment plot, feature extraction, initial clustering
November 3 – November 17	Mid-project check-in, trailerness model, refine clusters, build D3 views
November 17 – December 1	Refine D3 views, run case studies
December 1 – December 15	Final evaluation, presentation, report

6. References

D. M. Argaw, M. Soldan, A. Pardo, C. Zhao, F. Caba Heilbron, J. S. Chung, B. Ghanem. Towards Automated Movie Trailer Generation. CVPR, 2024. https://arxiv.org/abs/2404.03477

Q. Huang, Y. Xiong, A. Rao, J. Wang, D. Lin. MovieNet: A Holistic Dataset for Movie Understanding. ECCV, 2020. https://arxiv.org/abs/2007.10937

E. Fish, J. Weinbren, A. Gilbert. Rethinking Movie Genre Classification with Fine-Grained Semantic Clustering. ICIP, 2021. https://arxiv.org/abs/2012.02639

P. Xenopoulos, J. Rulff, L. G. Nonato, B. Barr, C. Silva. Calibrate: Interactive Analysis of Probabilistic Model Output. IEEE TVCG, 2023. https://arxiv.org/abs/2207.13770

J. Görtler, F. Hohman, D. Moritz, K. Wongsuphasawat, D. Ren, R. Nair, M. Kirchner, K. Patel. Neo: Generalizing Confusion Matrix Visualization to Hierarchical and Multi-Output Labels. CHI, 2022. https://arxiv.org/abs/2110.12536

S. M. Lundberg, S.-I. Lee. A Unified Approach to Interpreting Model Predictions. NeurIPS, 2017. https://arxiv.org/abs/1705.07874

R. Montalvo-Lezama, B. Montalvo-Lezama, G. Fuentes-Pineda. Improving Transfer Learning for Movie Trailer Genre Classification using a Dual Image and Video Transformer (Trailers12k). Information Processing & Management, 2023. https://arxiv.org/abs/2210.07983

M. Savardi, A. B. Kovács, A. Signoroni, S. Benini. CineScale: A Dataset of Cinematic Shot Scale in Movies. Data in Brief, 2021. https://doi.org/10.17632/th46h4vdwd.1